

UCT41042 - PRACTICAL FOR INTRODUCTION TO SMART SYSTEMS

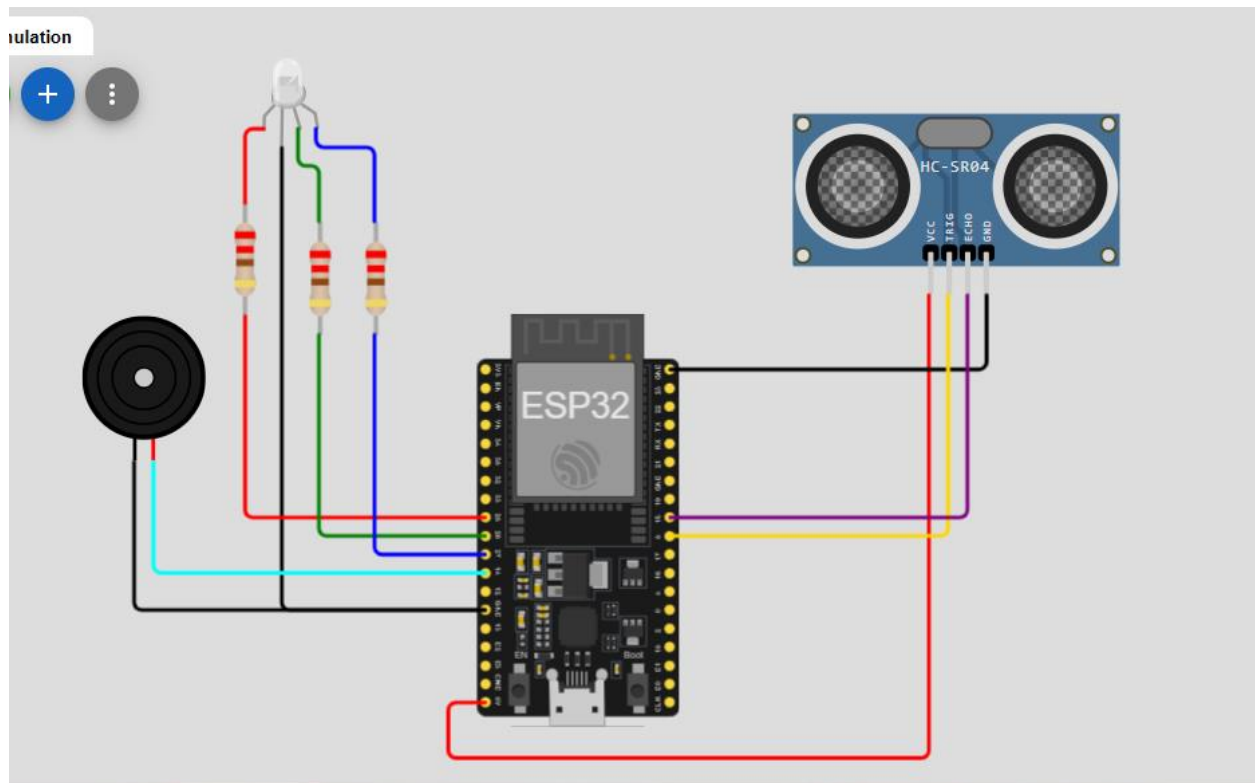
Continuous Assignment

Reg.No:SEU/IS/20/ICT/084

Date:2026/06/04

Q01.

Circuit design in Wokwi:



Code:

```
#define BLYNK_TEMPLATE_ID "TMPL6q3adVhDC"
#define BLYNK_TEMPLATE_NAME "Smart Parking Assistant"
#define BLYNK_AUTH_TOKEN "qewiSQH7C5NzYRaUZ18ZYXtl_wXvXZKv"

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>

// WiFi details
char ssid[] = "Wokwi-GUEST";
char pass[] = "";

// Sensor pins
#define TRIG_PIN 5
```

```
#define ECHO_PIN 18

// RGB LED pins
#define RED_LED 25
#define GREEN_LED 26
#define BLUE_LED 27

// Buzzer pin
#define BUZZER 14

void setup()
{
    // Start serial monitor
    Serial.begin(115200);

    // System start message
    Serial.println("SMART PARKING ASSISTANT");
    Serial.println("System Testing Started");

    // Set sensor pins
    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);

    // Set LED pins
    pinMode(RED_LED, OUTPUT);
    pinMode(GREEN_LED, OUTPUT);
    pinMode(BLUE_LED, OUTPUT);

    // Set buzzer pin
    pinMode(BUZZER, OUTPUT);

    // Connect to WiFi and Blynk
    Serial.println("Connecting...");
    Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

    // Check Blynk connection
    if (Blynk.connected())
    {
        Serial.println("Blynk Connected");
    }
    else
    {
        Serial.println("Blynk Not Connected");
    }
}
```

```
void loop()
{
    // Run Blynk
    Blynk.run();

    long duration;
    float distance;

    // Send ultrasonic signal
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);

    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);

    digitalWrite(TRIG_PIN, LOW);

    // Read echo signal
    duration = pulseIn(ECHO_PIN, HIGH);

    // Calculate distance
    distance = duration * 0.034 / 2;

    // Show distance
    Serial.print("Distance : ");
    Serial.print(distance);
    Serial.println(" cm");

    // Send distance to Blynk
    Blynk.virtualWrite(V0, distance);

    // Danger zone
    if (distance < 10){
        // Red LED ON
        digitalWrite(RED_LED, HIGH);
        digitalWrite(GREEN_LED, LOW);
        digitalWrite(BLUE_LED, LOW);

        // Fast buzzer
        digitalWrite(BUZZER, HIGH);
        delay(100);
        digitalWrite(BUZZER, LOW);
        delay(100);
    }
}
```

```

    // Update Blynk
    Blynk.virtualWrite(V1, 255);
    Blynk.virtualWrite(V2, 0);
    Blynk.virtualWrite(V3, 0);
    Blynk.virtualWrite(V4, "DANGER");

    Serial.println("Status : DANGER");
}

// Caution zone
else if (distance >= 10 && distance <= 30){
    // Yellow LED ON
    digitalWrite(REDF_LED, HIGH);
    digitalWrite(GREEN_LED, HIGH);
    digitalWrite(BLUE_LED, LOW);

    // Slow buzzer
    digitalWrite(BUZZER, HIGH);
    delay(500);
    digitalWrite(BUZZER, LOW);
    delay(500);

    // Update Blynk
    Blynk.virtualWrite(V1, 0);
    Blynk.virtualWrite(V2, 255);
    Blynk.virtualWrite(V3, 0);
    Blynk.virtualWrite(V4, "CAUTION");

    Serial.println("Status : CAUTION");
}

// Safe zone
else{
    // Green LED ON
    digitalWrite(REDF_LED, LOW);
    digitalWrite(GREEN_LED, HIGH);
    digitalWrite(BLUE_LED, LOW);

    // Buzzer OFF
    digitalWrite(BUZZER, LOW);

    // Update Blynk
    Blynk.virtualWrite(V1, 0);
    Blynk.virtualWrite(V2, 0);
    Blynk.virtualWrite(V3, 255);

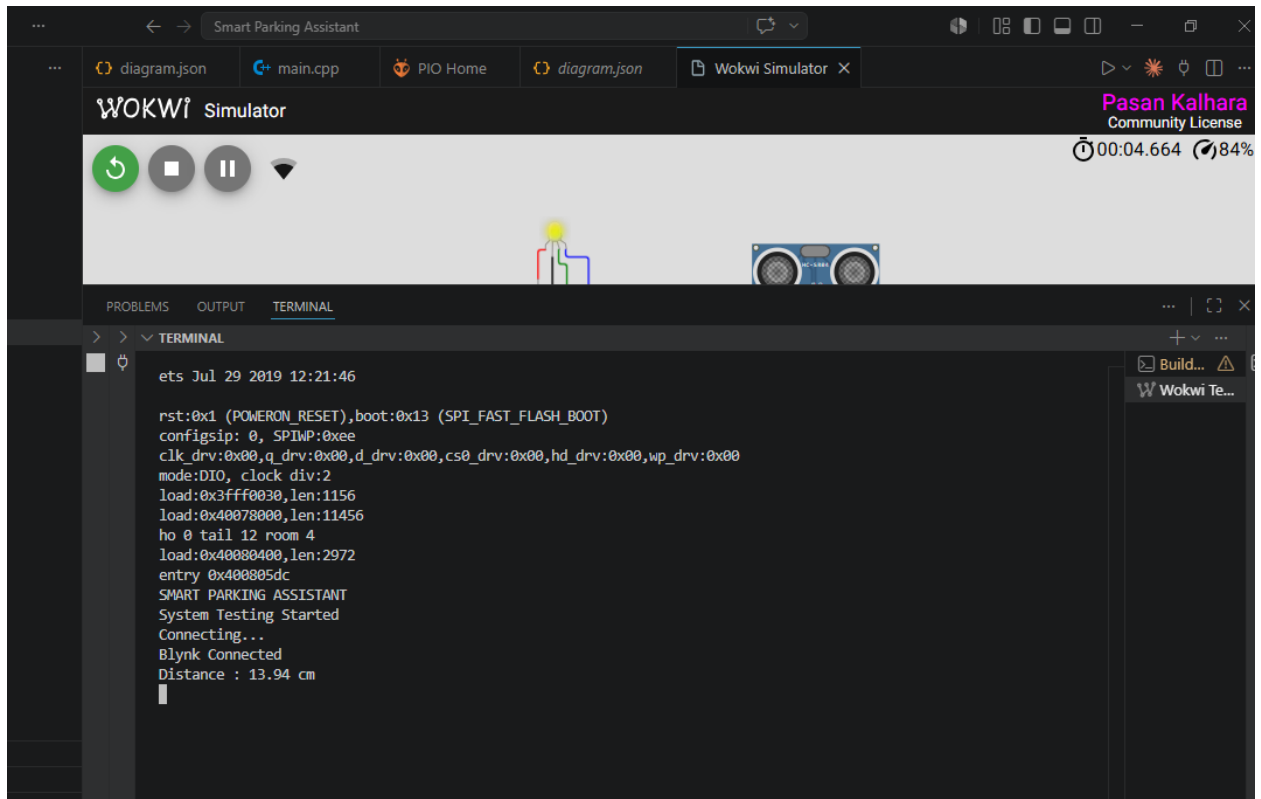
```

```
Blynk.virtualWrite(V4, "SAFE");

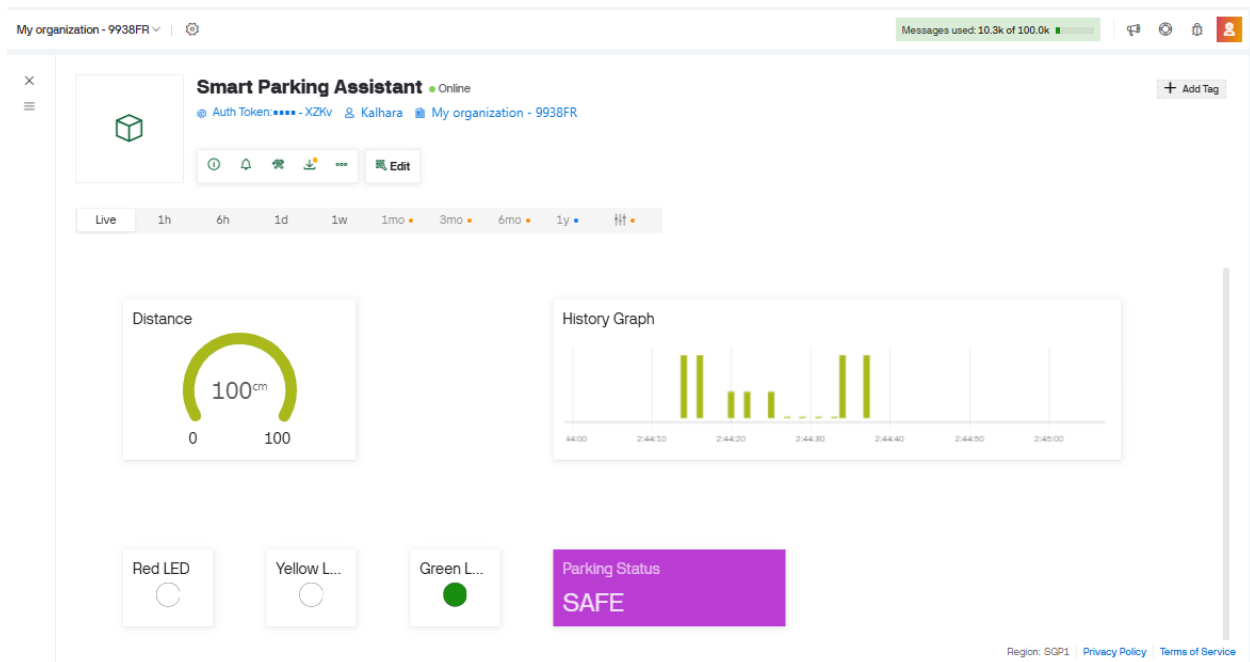
Serial.println("Status : SAFE");

delay(500);
}
}
```

Serial Monitor output:

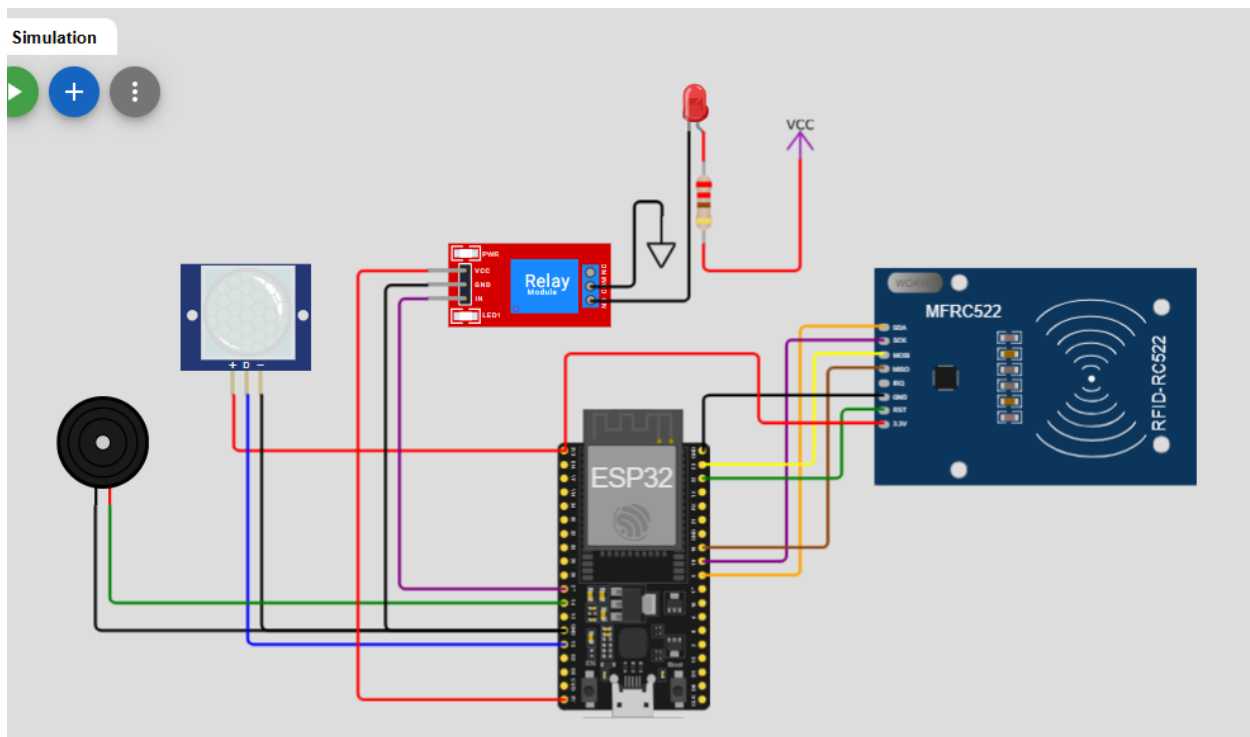


Blynk Web Dashboard:



Q02.

Circuit design in Wokwi:



Code:

```
#define BLYNK_TEMPLATE_ID "TMPL6Sc1JV5Xb"
#define BLYNK_TEMPLATE_NAME "Intrusion Detection System"
#define BLYNK_AUTH_TOKEN "rXIb-8A1eqDXVzU7wYKu-1nfId4IGYfI"

#include <Arduino.h>
#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
#include <SPI.h>
#include <MFRC522.h>

char ssid[] = "Wokwi-GUEST";
char pass[] = "";

#define PIR_PIN 13
#define BUZZER 14
#define DOOR_RELAY 27

#define SS_PIN 5
#define RST_PIN 22

MFRC522 rfid(SS_PIN, RST_PIN);

bool intrusion = false;
unsigned long intrusionTime = 0;

void setup()
{
    Serial.begin(115200);

    pinMode(PIR_PIN, INPUT);
    pinMode(BUZZER, OUTPUT);
    pinMode(DOOR_RELAY, OUTPUT);

    SPI.begin();
    rfid.PCD_Init();

    Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

    Serial.println("System Ready");
}

void loop()
```

```
{
  Blynk.run();

  int motion = digitalRead(PIR_PIN);

  if (motion == HIGH && intrusion == false)
  {
    intrusion = true;
    intrusionTime = millis();

    digitalWrite(BUZZER, HIGH);

    Serial.println("Intrusion Alert!");

    Blynk.virtualWrite(V0, 255);
  }

  if (intrusion == true)
  {
    if (rfid.PICC_IsNewCardPresent() &&
        rfid.PICC_ReadCardSerial())
    {
      String uid = "";

      for (byte i = 0; i < rfid.uid.size; i++)
      {
        uid += String(rfid.uid.uidByte[i], HEX);
      }

      Serial.print("Card UID: ");
      Serial.println(uid);

      // Replace with your RFID UID
      if (uid == "1234")
      {
        digitalWrite(DOOR_RELAY, HIGH);

        Blynk.virtualWrite(V1, "Access Granted");

        Serial.println("Access Granted");

        digitalWrite(BUZZER, LOW);

        intrusion = false;
      }
    }
  }
}
```



```

        delay(5000);

        digitalWrite(DOOR_RELAY, LOW);
    }
}

if (millis() - intrusionTime > 10000)
{
    Blynk.virtualWrite(V1, "Access Denied");

    Serial.println("Access Denied");

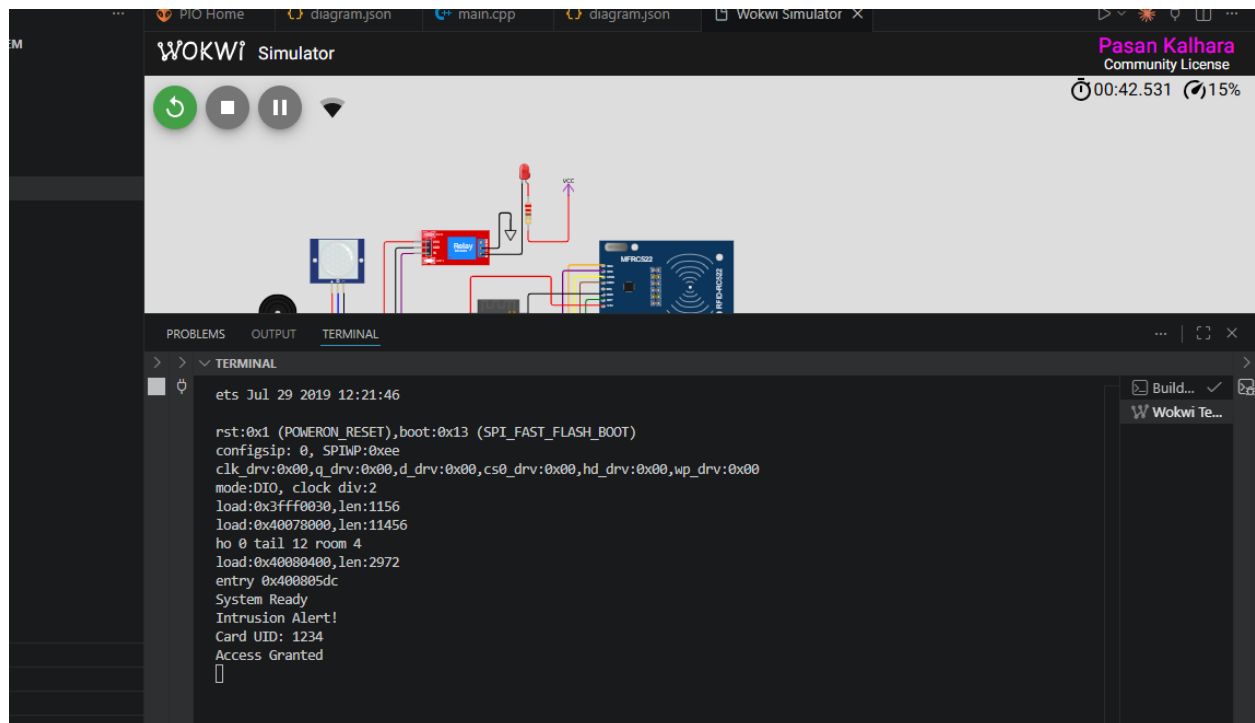
    digitalWrite(BUZZER, LOW);

    Blynk.virtualWrite(V0, 0);

    intrusion = false;
}
}
}
}

```

Serial Monitor output:



Blynk Web Dashboard:

The screenshot displays the Blynk Web Dashboard interface. On the left is a sidebar with navigation options: Get Started, Dashboards, Custom Data, Developer Zone, Devices (highlighted), Automations, Users, Organizations, Locations, Snapshots, Fleet Management, In-App Messaging, Predictive Maintenance, and Collapse. The main content area shows a dashboard titled 'Intrusion Detection System' with a status of 'Offline'. It includes a cube icon, a toolbar with icons for live, history, and settings, and a time range selector (Live, 1h, 6h, 1d, 1w, 1mo, 3mo, 6mo, 1y, All). Below the time range selector are two cards: 'Motion ...' with a blue circle icon and 'Access Status' with a green background and the text 'Access Granted'. The top right of the dashboard shows 'Messages used: 10.1k of 100.0k' and a user profile icon. The bottom right corner contains links for 'Region: EGP1', 'Privacy Policy', and 'Terms of Service'.

Blynk.Console My organization - 9938FR Messages used: 10.1k of 100.0k

Intrusion Detection System Offline + Add Tag

Auth Token: •••• - GYfi Kalhara My organization - 9938FR

Live 1h 6h 1d 1w 1mo 3mo 6mo 1y All

Motion ... Access Status
Access Granted

Region: EGP1 Privacy Policy Terms of Service